KATARZYNA SZYMANIAK

Machine Learning Research Scientist | Distribution Shift, Time Series and Multimodal Foundation Models

Edinburgh, UK github.com/KatarzynaSzymaniak katarzynaszymaniak.github.io

k.szymaniak.ai@gmail.com linkedin.com/in/katarzyna-szymaniak-ai Google Scholar

SUMMARY

Machine learning researcher (PhD, University of Edinburgh, 2026) specialising in **learning from noisy, non-stationary, multichannel time series** and making models that stay reliable under **distribution shift** and **scarce labels**. **Industry research:** Research Scientist Intern at Meta Reality Labs (CTRL-labs), working on multimodal sensor fusion and few-shot personalisation of a Conformer sequence model on high-rate wearable signals. **Methods:** domain generalisation to unseen regimes (EurIPS; extended version under review at TMLR), label-efficient active learning, self-supervised pretraining and parameter-efficient fine-tuning. **Rigour:** leakage-free benchmark design with nested cross-validation and non-parametric significance testing over $\sim$40k trained models; an open dataset in *Scientific Data* (Nature Portfolio). **Now: audio grounding** and **shortcut learning** in **multimodal LLMs** and **agents** (**counterfactual evaluation**, **RL post-training**), in collaboration with NVIDIA Research.

RESEARCH EXPERIENCE

Independent Research, in collaboration with NVIDIA Research Edinburgh, UK

Machine Learning Researcher, Audio Foundation Models for Understanding and Reasoning Aug 2026 – Present

- Investigating **audio grounding and shortcut learning in multimodal large language models (LLMs)**: whether models reason from the audio signal or from text priors and spurious acoustic cues.
- Designing **counterfactual evaluations and benchmarks** to measure robustness under distribution shift; planned next steps are RL post-training (GRPO, DPO, SFT with LoRA) to strengthen audio grounding and extension to agentic evaluation.

Meta Reality Labs (CTRL-labs) New York, NY

Research Scientist Intern, Continuous Navigation (Host: Fabio Stefanini) Sep 2022 – Mar 2023

- Extended the team's wrist-worn surface EMG (sEMG) decoding stack to **multimodal sEMG + IMU time-series fusion**: built the inertial data loading and preprocessing, trained **early-fusion neural decoders**, and diagnosed **modality imbalance** (one input dominating optimisation).
- Built an **end-to-end few-shot personalisation pipeline** for new custom gestures, from designing the human-subject data-collection protocol to per-user calibration of a **pretrained Conformer** (convolution-augmented Transformer) decoder.
- Evaluated **cross-session generalisation** after device re-wear with per-user **learning curves** over calibration budgets; presented results and limitations at the team's technical deep dive.
- Context: the team's cross-user sEMG decoding work was published in Nature (2025) and underpins the Meta Neural Band.

University of Edinburgh, School of Informatics Edinburgh, UK

Doctoral Researcher, Biomedical AI Sep 2020 – Jan 2026

- Led thesis research on why lab-trained sequence decoders fail in deployment (covariate shift from changing recording conditions and long-term drift), mitigating it with **domain generalisation, active learning and adaptive probabilistic modelling**.
- Owned the full research pipeline: experimental design, multi-session human-subject data collection (including amputee participants), signal processing, modelling and statistical evaluation.
- Built reusable data-processing and evaluation pipelines for multichannel signal studies, **adopted across the lab**.

SELECTED PUBLICATIONS

* equal contribution

Position Domain Generalisation in Myoelectric Control

K. Szymaniak, K. Nazarpour, H. Gouk *EurIPS 2025 Workshop | extended version under review at TMLR*

- Proposed **DR-cGMM**, a domain-regularised class-conditional Gaussian mixture model that generalises to unseen domains with zero data from them, via a covariance regulariser that shares structure across domains.
- Outperformed ERM, MMD and CORAL, **the only method to do so**; showed that popular deep domain generalisation methods gave no significant gain over ERM in this low-data regime.
- Designed a **leakage-free benchmark**: leave-one-domain-out model selection with nested cross-validation and Friedman/Nemenyi significance testing over **about 40,000 trained models**.

EMG Dataset for Gesture Recognition with Arm Translation (GREAT)

I. Kyranou*, K. Szymaniak*, K. Nazarpour | open dataset, 21,000+ accesses *Scientific Data (Nature Portfolio), 2025*

- Designed the acquisition protocol for an open benchmark of **16-channel sEMG at 2 kHz** with synchronised 18-sensor hand kinematics, recorded over a 3×3 grid of 9 arm positions to isolate a controlled distribution shift.
- Quantified the shift: **accuracy dropped by up to 10%** across positions, and position itself was **decodable from the signal (about 80%)**; showed that position-aware hierarchical decoding fails outside an i.i.d. setting (position encoder 58%), motivating the domain generalisation work above.

Recalibration of Myoelectric Control with Active Learning
K. Szymaniak, A. Krasoulis, K. Nazarpour *Frontiers in Neurorobotics, 2022*

- **Introduced active learning to myoelectric control for the first time**: the model queries labels only for windows ranked most informative (least-confidence, margin, entropy).
- Achieved **86% accuracy from 3.2 minutes** of actively queried data, versus 82% for both a full calibration session and random sampling (12 able-bodied participants; about 2 points gain in 2 transradial amputees; offline simulation).
- Identified a **failure mode of diversity-based batch selection**: it degraded margin sampling, likely because the most informative boundary samples cluster and are discarded as redundant.

Deep Feature Learning from Electromyographic Signals for Gesture Recognition Systems
W. Zhong, X. Jiang, K. Szymaniak, M. Jabbari, C. Ma, K. Nazarpour *IEEE TNSRE, 2026 (survey)*

EARLIER RESEARCH

- **Predicting behaviour from brain data** (University of Edinburgh, 2021; Prof. Oisin Mac Aodha, with the Centre for Discovery Brain Sciences): decoded the location of freely behaving mice from in vivo deep-brain calcium imaging, using class-activation-style attribution to identify the neuronal populations driving predictions.
- **Sequence models for EEG** (Swansea University, 2020; Dr Jingjing Deng, Prof. Xianghua Xie): built deep sequential models for EEG in decision deadlocks, handling temporal dependencies, non-stationarity, cross-channel connectivity and noise, towards brain–computer interfaces.
- **HoloLens neuroimaging for medical education** (Swansea University, 2019; Best Student Project award): deployed 3D MRI brain meshes on Microsoft HoloLens (Unity) as a mixed-reality teaching tool.

EDUCATION

PhD, Biomedical Artificial Intelligence, University of Edinburgh, UK Sep 2020 – Jan 2026
Thesis: Mitigating Confounding Factors in Myoelectric Control Through Adaptive Modelling and Learning. Supervisors: Prof. Kianoush Nazarpour, Prof. Timothy Hospedales.

MRes, Biomedical Artificial Intelligence, University of Edinburgh, UK Sep 2019 – Sep 2020
MRes, Visual Computing, Swansea University, UK Sep 2018 – Sep 2020
BSc (Hons) Computer Science, First Class Honours, Swansea University, UK Sep 2015 – Jul 2018
Oxford Machine Learning Summer School (OxML), University of Oxford, UK Aug 2022

TECHNICAL SKILLS

Machine Learning: deep learning, sequence models (Transformers, Conformers, CNN/TCN, RNN/LSTM), time-series classification, self-supervised and representation learning, transfer learning, domain adaptation and generalisation, OOD generalisation, few-shot and active learning, multimodal fusion, uncertainty estimation, interpretability

LLMs and Post-Training: SFT, GRPO, PEFT/LoRA, LLM and agent evaluation, counterfactual evaluation

Statistics and Evaluation: experimental design, benchmark design, model selection, hyperparameter tuning, cross-validation, significance testing

Signal Processing: filtering, spectral and time-domain features, real-time windowing; EMG, IMU, EEG, audio

Programming and MLOps: Python, PyTorch, PyTorch Lightning, Hydra, vLLM, scikit-learn, NumPy, SciPy, Pandas, SQL, MATLAB, Java, TensorFlow; MLflow, Weights & Biases, Docker, Kubernetes, Git

TEACHING

Teaching Assistant, Machine Learning Practical (MLP), University of Edinburgh Sep 2024 – Jun 2025
Teaching Assistant, Entrepreneurship and Innovation, University of Edinburgh Sep 2023 – Jul 2024
Mentored student teams building ML-driven MVPs for real-world health and technology problems.
Lab Demonstrator, Java OOP, Data Structures and Algorithms, Swansea University Jan 2018 – Sep 2019

SERVICE AND LEADERSHIP

- Peer Reviewer, IEEE TNSRE; Frontiers in Neurorobotics Nov 2024 – Aug 2025
- Co-organiser, Women in Machine Learning (WiML) Workshop at NeurIPS 2021 Dec 2021
- Co-organiser, two neuroprosthetics co-creation and co-design workshops, University of Edinburgh 2021, 2022
- Research collaborations: Centre for Discovery Brain Sciences, Edinburgh (2020); CUBRIC, Cardiff (2018–2020)